

REVISION SERIES(Differentiability+Continuity)

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Practice Question 1

If

$$y = (\tan^{-1} x)^2,$$

show that

$$(1 + x^2)^2 \frac{d^2 y}{dx^2} + 2x(1 + x^2) \frac{dy}{dx} = 2.$$

Practice Question 2

A right circular cone has a fixed slant height.

Show that the cone of maximum volume has semi-vertical angle

$$\theta = \tan^{-1}(\sqrt{2}).$$

Detailed Solutions

Solution of Question 1

Given,

$$y = \left(\tan^{-1} x\right)^2.$$

Differentiate:

$$\frac{dy}{dx} = \frac{2 \tan^{-1} x}{1 + x^2}.$$

Again,

$$\frac{d^2 y}{dx^2} = \frac{2 - 4x \tan^{-1} x}{(1 + x^2)^2}.$$

Hence,

$$(1 + x^2)^2 \frac{d^2 y}{dx^2} = 2 - 4x \tan^{-1} x.$$

Also,

$$2x(1 + x^2) \frac{dy}{dx} = 4x \tan^{-1} x.$$

Adding,

$$(1 + x^2)^2 \frac{d^2 y}{dx^2} + 2x(1 + x^2) \frac{dy}{dx} = 2.$$

Hence proved.

$$\boxed{(1 + x^2)^2 y'' + 2x(1 + x^2) y' = 2.}$$

Solution of Question 2

Let

$$r = l \sin \theta, \quad h = l \cos \theta,$$

where l is the fixed slant height.

Therefore,

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi l^3 \sin^2 \theta \cos \theta.$$

Differentiate:

$$\frac{dV}{d\theta} = \frac{1}{3}\pi l^3 \sin \theta (2 \cos^2 \theta - \sin^2 \theta).$$

For maximum volume,

$$\frac{dV}{d\theta} = 0.$$

Ignoring the trivial solution,

$$2 \cos^2 \theta = \sin^2 \theta.$$

Hence,

$$\tan^2 \theta = 2.$$

Therefore,

$$\theta = \tan^{-1}(\sqrt{2}).$$

Hence proved.

Excellent! Keep Practising Mathematics Every Day.